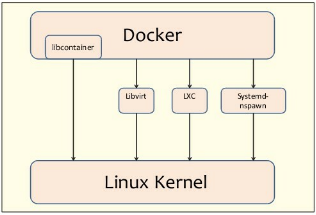

Figure 1: Docker container

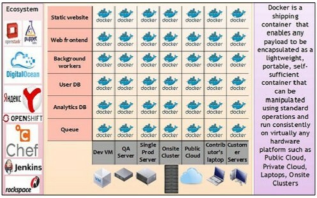

Figure 2: Use cases and eco-system of docker

user IDs. The kernel's *cgroups* provides isolation from the network, memory, CPU and block I/O. In some versions of Docker, the *libcontainer* library directly uses the facilities for virtualisation provided by the kernel. It also uses abstracted virtualisation interfaces via *LXC*, *system-nspawn* and *libvirt*.

Docker enables applications to be assembled from components and eliminates the friction between the production environment, QA and development. By using Docker, we can create and manage containers; it enables us to create highly distributed systems by allowing workers' tasks, multiple applications and other processes. Essentially, it provides a Platform as a Service (PaaS) style of deployment and, hence, enables IT to ship faster and run the same application unchanged on the data centre, VMS, the cloud and on laptops.

Docker can be integrated into various infrastructure automation or configuration management tools (such as Chef, CFEngine, Puppet and Salt), in continuous integration tools, in cloud service providers such as Amazon Web Services, private cloud products like OpenStack Nova, and so on.

Docker installation

Docker can be installed on various operating systems such as Microsoft Windows, CentOS, Fedora, SUSE, Amazon EC2, Ubuntu, and so on. We will be covering installation for three operating systems—Microsoft Windows, CentOS and Amazon EC2.

On Windows

As the Docker engine uses Linux kernel features, to execute it on Windows, you need to use a virtual machine. A Windows Docker client is used to control the virtualised Docker engine to execute, manage and build Docker containers.

Docker has been tested on Windows 7.1 and 8, apart from other versions. The processor needs to support hardware virtualisation. To make the process easier, an application called Boot2Docker has been designed that installs the virtual machine and runs the Docker.

1. First, download the latest version of Docker for the Windows operating system.
2. Follow the steps to complete the installation, which will result in the installation of Boot2Docker, Boot2Docker Linux ISO, VirtualBox, Boot2Docker management tool and MSYS-git.
3. After this, run Boot2Docker to start the shell script or execute *Program Files > Boot2Docker. Start script* will ask you to enter an ssh key passphrase or simply press *[Enter]*. This script will connect to a shell session in the virtual machine. According to requirements, it will initialise a new VM virtual machine and start it.

To upgrade, download the latest version of Docker for the Windows operating system, and run the installer to update the Boot2Docker management tool. To upgrade the existing virtual machine, open a terminal and perform the following steps:

```
boot2docker stop
boot2docker download
boot2docker start
```

On CentOS

Docker is available by default in the CentOS-Extras repository on CentOS 7. Hence, to install it, run *sudo yum install docker*. In Centos-6.5, the Docker package is part of *Extra Packages* for the Enterprise Linux (EPEL) repository.

On Amazon EC2

Create an AWS account. To install Docker on AWS EC2, use

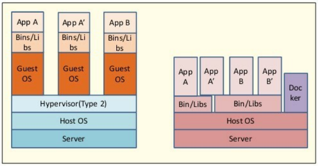

Figure 3: Comparing virtual machines and Docker (source: Docker website)